

‘EXPERIMENT 3

EXPLORATORY DATA ANALYSIS (EDA) – DATA CLEANING AND DATA TRANSFORMATION

Aim

To identify and handle missing values, duplicate records, outliers, inconsistent data, and perform data transformation techniques such as normalization and encoding using Python Pandas.

Algorithm 1

Identifying Missing Values

Step 1

Import pandas library

Step 2

Load dataset

Step 3

Use isnull() function

Step 4

Count missing values

Step 5

Display result

Program 1

```
import pandas as pd

df=pd.read_csv("C:/Users/91754/Downloads/transactions.csv")
print(df.head())
```

	Transaction ID	Timestamp	Sender Name	\
0	4d3db980-46cd-4158-a812-dcb77055d0d2	2024-06-22 04:06:38	Tiya Mall	
1	099ee548-2fc1-4811-bf92-559c467ca792	2024-06-19 06:04:49	Mohanlal Bakshi	
2	d4c05732-6b1b-4bab-90b9-efe09d252b99	2024-06-04 04:56:09	Kismat Bora	
3	e8df92ee-8b04-4133-af5a-5f412180c8ab	2024-06-09 09:56:07	Ayesha Korpai	
4	e7d675d3-04f1-419c-a841-7a04662560b7	2024-06-25 08:38:19	Jivin Batta	

	Sender UPI ID	Receiver Name	Receiver UPI ID	Amount (INR)	\
0	4161803452@okaxis	Mohanlal Golla	7776849307@okybl	3907.34	
1	8908837379@okaxis	Mehul Sankaran	7683454560@okaxis	8404.55	
2	4633654150@okybl	Diya Goel	2598130823@okicici	941.88	
3	7018842771@okhdfcbank	Rhea Kothari	2246623650@okaxis	8926.00	
4	1977143985@okybl	Baiju Issac	5245672729@okybl	2800.55	

	Status
0	FAILED
1	SUCCESS
2	SUCCESS
3	SUCCESS
4	SUCCESS

```
print(df.isnull())
```

	Transaction ID	Timestamp	Sender Name	Sender UPI ID	Receiver Name	\
0	False	False	False	False	False	
1	False	False	False	False	False	
2	False	False	False	False	False	
3	False	False	False	False	False	
4	False	False	False	False	False	
..	...	...	...	...	...	
995	False	False	False	False	False	
996	False	False	False	False	False	
997	False	False	False	False	False	
998	False	False	False	False	False	
999	False	False	False	False	False	

	Receiver UPI ID	Amount (INR)	Status
0	False	False	False
1	False	False	False
2	False	False	False
3	False	False	False
4	False	False	False
..	...	...	...
995	False	False	False
996	False	False	False
997	False	False	False
998	False	False	False
999	False	False	False

[1000 rows x 8 columns]

Program 2

Count Missing Values

```
print(df.isnull().sum())
```

```
Transaction ID    0
Timestamp         0
Sender Name       0
Sender UPI ID     0
Receiver Name     0
Receiver UPI ID   0
Amount (INR)      0
Status            0
dtype: int64
```

Algorithm 2

Replacing Missing Values

Step 1

Calculate mean value

Step 2

Replace null values

Step 3

Verify replacement

Program 3

```
df["Amount (INR)"] = df["Amount (INR)"].fillna(df["Amount (INR)"].mean())

df["Sender Name"] = df["Sender Name"].fillna("Unknown")
df["Sender UPI ID"] = df["Sender UPI ID"].fillna("Unknown")
df["Receiver Name"] = df["Receiver Name"].fillna("Unknown")
df["Receiver UPI ID"] = df["Receiver UPI ID"].fillna("Unknown")
df["Status"] = df["Status"].fillna("Unknown")

print(df)
```

	Transaction ID	Timestamp \
0	4d3db980-46cd-4158-a812-dcb77055d0d2	2024-06-22 04:06:38
1	099ee548-2fc1-4811-bf92-559c467ca792	2024-06-19 06:04:49
2	d4c05732-6b1b-4bab-90b9-efe09d252b99	2024-06-04 04:56:09
3	e8df92ee-8b04-4133-af5a-5f412180c8ab	2024-06-09 09:56:07
4	e7d675d3-04f1-419c-a841-7a04662560b7	2024-06-25 08:38:19
...	...	...
995	843d87b5-902e-4336-9438-aa76d74e13e4	2024-07-03 23:15:06
996	2dbbb577-4617-44b1-9a51-5fdf18ebb2ab	2024-06-15 15:42:57
997	7c38e684-5968-432e-8acb-5bbe0f718fad	2024-06-18 17:27:37
998	27624b55-fd6d-45ce-bbc5-0c2958d48865	2024-06-05 21:57:36
999	b792bf64-2047-4435-aac1-27ee95b134d6	2024-06-21 15:52:40

Algorithm 3

Removing Duplicate Records

Step 1

Use duplicated()

Step 2

Count duplicate records

Step 3

Remove duplicates

Step 4

Verify dataset

Program 4

```

print("Duplicate Records:")
print(df.duplicated())
print("Total Duplicates:", df.duplicated().sum())
df.drop_duplicates(inplace=True)
print("Dataset after removing duplicates")
print(df)

```

Duplicate Records:

```

0      False
1      False
2      False
3      False
4      False

```

```

...
995     False
996     False
997     False
998     False
999     False

```

Length: 1000, dtype: bool

Total Duplicates: 0

Dataset after removing duplicates

	Transaction ID	Timestamp \
0	4d3db980-46cd-4158-a812-dcb77055d0d2	2024-06-22 04:06:38
1	099ee548-2fc1-4811-bf92-559c467ca792	2024-06-19 06:04:49
2	d4c05732-6b1b-4bab-90b9-efe09d252b99	2024-06-04 04:56:09
3	e8df92ee-8b04-4133-af5a-5f412180c8ab	2024-06-09 09:56:07
4	e7d675d3-04f1-419c-a841-7a04662560b7	2024-06-25 08:38:19
..	...	...
995	843d87b5-902e-4336-9438-aa76d74e13e4	2024-07-03 23:15:06
996	2dbbb577-4617-44b1-9a51-5fdf18ebb2ab	2024-06-15 15:42:57
997	7c38e684-5968-432e-8acb-5bbe0f718fad	2024-06-18 17:27:37
998	27624b55-fd6d-45ce-bbc5-0c2958d48865	2024-06-05 21:57:36
999	b792bf64-2047-4435-aac1-27ee95b134d6	2024-06-21 15:52:40

Algorithm 4

Detecting Outliers

Step 1

Calculate Quartiles

Step 2

Compute IQR

Step 3

Calculate lower and upper limits

Step 4

Identify outliers

Program 5

```
Q1 = df["Amount (INR)"].quantile(0.25)
```

```
Q3 = df["Amount (INR)"].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
lower = Q1 - 1.5 * IQR
```

```
upper = Q3 + 1.5 * IQR
```

```
outliers = df[(df["Amount (INR)"] < lower) |  
              (df["Amount (INR)"] > upper)]
```

```
print(outliers)
```

Empty DataFrame

Columns: [Transaction ID, Timestamp, Sender Name, Sender UPI ID, Receiver Name, Receiver UPI ID, Amount (INR), Status]

Index: []

Program 6- Normalisation

```
from sklearn.preprocessing import MinMaxScaler
```

```
scaler = MinMaxScaler()
```

```
df[["Amount (INR)"]] = scaler.fit_transform(df[["Amount (INR)"]])
```

```
print(df[["Amount (INR)"]])
```

	Amount (INR)
0	0.389262
1	0.840584
2	0.091661
3	0.892914
4	0.278189
..	...
995	0.927452
996	0.225781
997	0.484656
998	0.948078
999	0.140306

[1000 rows x 1 columns]

Program 7- Label Encoding

```

from sklearn.preprocessing import LabelEncoder

encoder = LabelEncoder()

df["Status"] = encoder.fit_transform(df["Status"])

print(df)

```

	Transaction ID	Timestamp	\
0	4d3db980-46cd-4158-a812-dcb77055d0d2	2024-06-22 04:06:38	
1	099ee548-2fc1-4811-bf92-559c467ca792	2024-06-19 06:04:49	
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3	e8df92ee-8b04-4133-af5a-5f412180c8ab	2024-06-09 09:56:07	
4	e7d675d3-04f1-419c-a841-7a04662560b7	2024-06-25 08:38:19	
..	...	...	
995	843d87b5-902e-4336-9438-aa76d74e13e4	2024-07-03 23:15:06	
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999	b792bf64-2047-4435-aac1-27ee95b134d6	2024-06-21 15:52:40	

	Sender Name	Sender UPI ID	Receiver Name	\
0	Tiya Mall	4161803452@okaxis	Mohanlal Golla	
1	Mohanlal Bakshi	8908837379@okaxis	Mehul Sankaran	
2	Kismat Bora	4633654150@okybl	Diya Goel	
3	Ayesha Korpai	7018842771@okhdfcbank	Rhea Kothari	
4	Jivin Batta	1977143985@okybl	Baiju Issac	
..	...	...	...	
995	Anay Karan	7041132054@okhdfcbank	Dhanush Salvi	
996	Nirvaan Aurora	1200592114@okybl	Faiyaz Kara	
997	Bhamini Srinivasan	6922445535@okicici	Mishti Kala	
998	Krish Kohli	4638471232@okhdfcbank	Sahil Bail	
999	Diya Trivedi	0385561234@oksbi	Dhanuk Mane	

Program 8

One-Hot Encoding

```

status_encoded = pd.get_dummies(df["Status"])

print(status_encoded)

```

	0	1
0	True	False
1	False	True
2	False	True
3	False	True
4	False	True
..	...	...
995	False	True
996	True	False
997	True	False
998	True	False
999	False	True

Result

Thus, missing values, duplicate records, and outliers were identified and handled successfully. Data transformation techniques such as normalization and encoding were performed, and the dataset was prepared for further exploratory data analysis and machine learning applications.